

Towards Prediction of Power Consumption of Virtual Machines for Varying Loads

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Abstract—Power management and load balancing in data centers are becoming critical with the growing size of the infrastructure. In front of an increasing number of cloud networks, virtual machines (VMs) and containers, smart management and control decisions are required, in order to instantiate or mobilize these virtual components. As regards reducing power consumption, it is also essential to consolidate virtual resources on the minimum possible number of servers compatible with performance requirements. In such a dynamic scenario, estimating the power that can be ascribed to a specific virtual component and its relation with the offered workload could be beneficial for optimized resource scheduling. Predicting the power consumption caused by a specific virtual component is however challenging. In this paper, two power models that relate a VM workload with the fraction of power consumed attributable to the VM are developed based on power profiling of a server. The accuracy of the proposed models is improved approximately by 3% compared to other existing models.

Index Terms—Virtual machine, power modeling, regression model, power profiling

I. INTRODUCTION

Measuring and controlling power consumption in data centers has gained increasing relevance with the advent of cloud computing [1]. With advanced processors and parallel processing, measuring the power that can be attributed to the execution of a process or application has become more challenging [2]. In fact, though many power models and power meters are available to provide precise power measurements for hardware devices, determining the fraction of energy pertaining to the execution of virtualized entities, like VMs or containers is still challenging [3]. In the modern world, the emergence of cloud computing and of "softwarized" networks has led to a massive increase in virtualization. Regarding data centers, estimates of 2008-2010 already reported an overall energy requirement of about 100 billion kWh per year and an energy expenditure of global companies of about 40 billion US dollars [4].

An intrinsic difficulty in measuring the power consumption of a virtual component precisely is due to the fact that available power meters cannot be connected directly to the virtual component to perform the measurement [5]. On the other hand,

accurate power measurements of individual virtual components can help improve the overall energy efficiency. Accuracy in power measurements can allow obtaining energy-efficient load balancing algorithms and effective server consolidation [6]. It is also important to charge customers with a fair amount for the resources they consume [7], which would be made possible by more accurate power measurements.

In this paper, a VM's attributed power is predicted with a model which is trained by the server power behavior with varying VM workload. The server power behavior is monitored with respect to the instructions requested or processed by the individual virtual machine. With this study, a relation between the change in a specific VM's instruction execution rate (in units of Million Instructions Per Second – MIPS) and the change in physical server power (Watts – W) can be established, which helps us in modeling the power required by the execution of a particular workload on an individual VM. A state of the art of the power modeling of virtualized instances is presented in Section II and our methodology is introduced in Section III. Section IV describes the experimental test bed, while results and their discussion are presented in Section V. Section VI contains the conclusions.

II. LITERATURE REVIEW

Power models available for physical devices cannot be mapped directly to Virtual Machines. Indeed, it is the hardware that consumes power, and the influence of a virtual entity in this process is mediated by many factors (presence of the hypervisor, number of virtual entities sharing the resources and their scheduling, etc.). Challenges encountered include power measurement at the VM level, VM resource management, load balancing and VM power billing and budgeting [4]. Power optimization of virtual instances in one way can be optimized by using effective scheduling algorithms. The scheduling of dynamic tasks among VMs based on Weighted Round Robin is proposed in [8]. Round Robin with its numerous variations has been used in most of the scheduling processes in virtual environments. The algorithm presented in [8] is a combination of

classical Round Robin and Weighted Round Robin. They use the capacity, task length, priority and load on VMs to allocate tasks to the appropriate VMs. To optimize the performance of the system with balanced load, the processing time of each task on a VM is estimated and must be minimized. A limitation of the above-mentioned methods is that they measure the capacity of the whole system and not the resource utilization attributable to individual VMs. To deal with the dynamic load and management of resources in virtual networks, a discrete state transition model is designed in [9]. This system provides a predictive controller for virtual systems based on a non-linear power model. The states are defined based on the server state (on or off), the number of VMs and their virtual clock. The processor can switch among these states during run-time to maintain the workload among different virtual machines. This model, however, has the limitation of being unsuitable for a large dimension of the state space, as this will make scheduling and design complex.

In any case, for power-efficient load balancing and scheduling, the management and control entities should be able to know the actual power consumed by individual (real or virtual) components. Precise measurements of resource utilization and performance metrics of virtual machines are important for optimization [10]. To measure the power at the virtual level, many of the power models available use the hardware resource consumption of the whole system [11]. There is no accurate power meter available to measure the consumption of a single core of a server or for a single VM [5].

Several methods of monitoring VM performance are available either from inside a VM or from outside of it [4]. The methods that monitor the VM performance from outside are known as black box methods, and use different software and hardware power meters. On the other hand, white box methods use a proxy program inside the VM to collect its performance metrics. This approach does not completely preserve the integrity of the VM or client and is not preferred in many cases. According to [4], black box methods using software meters or hardware counters are more reliable. However, they involve anyway a little overhead and their own effect on the performance of the system is not considered in most of the studies. Different software tools such as OProfile, PERF, Sysstat, Pfmmon and many others are available for observing different resources like memory access, storage and other performance measurements [4]. The most common approach towards modeling VM power consumption is in the adoption of linear models such as Least Squares Estimation (LSE), Lasso, Mantis, etc. The non-Linear models used in most cases are Polynomial Regression, Multi-Gaussian Regression and Exponential Regression [4].

Several power models for virtual machines have been developed using different power monitoring tools and modeling approaches. A power model in [12] was designed to estimate the power consumed by the physical host when changing the number of virtual machines running on the server. The power profile of a server was obtained for the number of VMs running and type of load (either CPU-only or network-

only) on them. Experiments show that power consumption increased non-linearly by increasing the number of VMs, and this increase is significant for network-only VMs. LLSR was applied to the observed data, which shows the logarithmic relation between VM resource utilization and host power. However, it is difficult to estimate the consumption of an individual VM when more than one VM is running on a host, and only the aggregated virtual load is considered in [12].

Another study was conducted to evaluate the power models of processors, VMs and servers [11]. According to their survey, most of the power models of the virtual machines use physical machine counters and apply different linear and non-linear regression models to estimate the power consumption of the VM. A combined power model for a VM is presented in this survey, derived through the combination of two different models for the same VM but for different traffic types [13]. First, a CPU-intensive workload and then a network-intensive workload were used, with the adoption of linear models. A combined power model for the VM is then proposed using the weighted values for each of these units. Another study examined in this survey suggested that a different cache level leads to a different amount of power consumption. This metric was incorporated in the research work to develop the power model of a VM by observing the level of memory hit by the process. Different power bounds were created by observing power at different levels and these bounds are then used to estimate the power of processes accessing memory [14]. Another proposition is to use hardware counters. In this respect, the concept of binning was adopted corresponding to the CPU utilization. The bins contained the quantized values of the performance counters, which were used in the model by means of a non-linear Gaussian Mixture Vector Quantization Technique (GMVQ) to find the optimized vector values [15]. The prediction error of this model was found to be around 10%.

In [16] a modified system architecture to collect the resource consumption of VMs was proposed. The power model was then developed by using the Performance Monitoring Counter (PMC) for the VMs and dependencies of these PMCs on each other. A non-linear regression model, Support Vector Regression (SVR), was used to develop the power model for this system. The study shows that different hardware PMCs are highly correlated and incorporating their relation in the model can increase the accuracy of power resource measurements. This model has a prediction error of around 8%.

III. METHODOLOGY

In this paper, a power model for virtual machines is developed by using the relation between the VM workload and the power consumed at the server. As the load on the virtual or physical machine corresponds to the number of instructions requested or processed by the machine, the number of instructions that a VM processed during the experiment is monitored. In the literature, performance monitoring counters (PMC) are already used, but those counter values were extracted for the

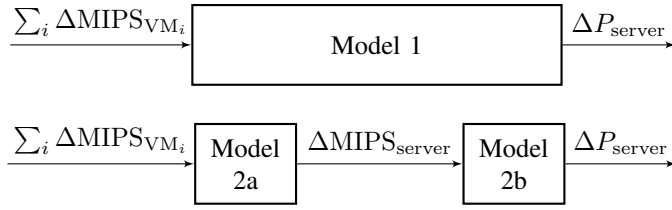


Fig. 1. Illustration of the two strategies to model power change from the change in virtual machine MIPS.

whole server [13]. In this paper PMCs for each VM are monitored individually and the power relation corresponding to the VM's instructions is modelled. Instructions of a particular VM are also monitored at the physical machine. The instruction count changes by increasing the workload on a similar virtual machine, and also if other virtual machines are running in parallel. This approach of measuring instruction counts at VM level helps in obtaining the power model for a VM in particular. The change in instructions' rate is calculated by using Equations (1) and (2), where i is the number of VMs running and j represents the vCPU utilization (or workload on the VM). Furthermore, the power profile of the server is obtained by monitoring its power, and the change in power is obtained using Equation (3).

$$\Delta\text{MIPS}_{\text{VM}_i} = \text{MIPS}_{\text{VM}_i,j} - \text{MIPS}_{\text{VM}_i,\text{idle}} \quad (1)$$

$$\Delta\text{MIPS}_{\text{server}} = \text{MIPS}_{\text{server},j} - \text{MIPS}_{\text{server},\text{idle}} \quad (2)$$

$$\Delta P_{\text{server}} = P_{\text{server},j} - P_{\text{server},\text{idle}} \quad (3)$$

where, $i \in \{1, 2\}$ and $j \in \{10, 20, \dots, 100\}$.

To determine the change in power and instructions, idle values for both variables were also obtained. Idle server power is found to be 17W for the machine whose characteristics are reported in Section IV, and MIPS at the server when idle were 0.4 MIPS. Also, idle MIPS at each VM observed was in the range from 0.3 MIPS to 0.7 MIPS. Two different modeling methods are used for the data sets, which will be referred to as Model1 and Model2 in this paper. In Model1, a data set containing the instruction count collected at the VM level is used. The sum of the increase in instruction rates from idle level of all VMs running the specific experiment, $\sum_i \Delta\text{MIPS}_{\text{VM}_i}$, will be fed as the predictor to Model1. The response variable for Model1 is the change in power at the server corresponding to the varying load. For training of Model2, the model is divided into two sub-models, which will be referred to as Model 2a and Model 2b. In model 2a, the relation between change in MIPS at a VM and the change in MIPS at the server is determined, whereas, in model 2b, the server power change, ΔP_{server} , is modeled corresponding to the change in instructions' rate of the VMs running on the server. Figure 1 illustrates the considered modeling approaches. Further, each modeling method is trained with two regression models: Linear regression and linear SVR, which will be further referred to as LLSR (Linear Least Squares

Regression) and LSVR (Linear Support Vector Regression), respectively. Prediction accuracy and error for both models are compared in the results section.

A. Linear Least Squares Regression

LLSR model is the simplest and most commonly used predictive analysis model that represents the linear relationship between a single dependent variable y and a single (or many) independent variable(s) $\mathbf{x} = [x_1, x_2, \dots, x_n]$ [17]. Instead of using absolute difference values, LLSR uses the square values of the differences between measured and predicted points. The sum of square distance between the data points and predicted values should be minimum to get the best approximate value from the model. A simple LLSR model with single dependent and single independent variable can be written as Equation (4),

$$y = \beta_0 + \beta_1 x + \epsilon \quad (4)$$

In Equation (4), β relates the change in predictor variable x to the mean of the response variable y , and ϵ is the random noise error. The output function of this model is the best fitted line for the given data points.

B. Support Vector Regression

The Support Vector Machine (SVM) is a supervised learning algorithm which is being used for classification and regression in datasets. SVM is useful for datasets where prediction of response is difficult due to random system response [18]. For this experiment, the mapping of "virtual" instructions of a VM on the physical machine's instructions is not straightforward, as it can change dynamically. So, the effect of VM instructions on the physical device and consequently on the power consumption of the server is analyzed using the SVM regression model. To study in depth the effect of one variable on the other, SVM maps data points of the training set into multi-dimensional feature space using some fixed mapping. The hidden relation among these data at higher dimension space is then determined and a linear model is constructed in this feature space. The so-called primal formula for linear SVM to derive the vector β of the coefficients of the linear regression model through (constrained) minimization is

$$f(\beta) = \frac{1}{2} \beta' \beta + C \sum_{n=1}^N (\epsilon_n + \epsilon_n^*) \quad (5)$$

where prime denotes transpose. The quality of the estimated values is measured with a loss function, for which slack variables are introduced. The slack variables ϵ_n and ϵ_n^* defines the allowed range of error. The data points outside this range are penalized. This penalty in the model is determined by the constant C , a positive numeric value that controls the penalty imposed on observations that lie outside the ϵ margin.

IV. TEST-BED

The experimental test-bed has been set up by using the OpenStack software. "Intel(R) Core(TM) i7-6770HQ" CPU is used as the server, whereas "compute (NOVA)" serves as

TABLE I
SYSTEM CONFIGURATION FOR THE SERVER AND VM

	Server	VM
Processor	Intel(R) Core(TM) i7-6770HQ CPU	Intel(R) Core(TM) i7-6770HQ CPU
No. of CPUs	8	1
OS	Linux 4.9.0-3-amd64	Linux 4.4.0-134-generic
CPU frequency	2.60GHz	2.592GHz
Freq scaling	1.099GHz-2500GHz	-
Cache	6144KB	16384KB
Memory	32GB	2GB
Hypervisor	Nova (qemu-KVM)	-

the hypervisor. The server has five VMs, all with similar configuration. Further configuration details about the server and VMs are listed in Table I.

The program “stress-ng” is used to generate the workload on VMs. The selected stress methods with integer and floating point operations runs on the VM in sequence. For these experiments only CPU-intensive workload is considered, where the workload is increased by increasing the virtual CPU (vCPU) utilization on the VM. In each experiment workload on a VM is increased by 10% of virtual CPU utilization and goes up to 100%. The power behavior of the server is also observed in different scenarios, such as no load, single VM run, 2 VMs running. The virtual machine instructions processed by a specific VM are measured using the Linux PERF tool inside the VM. The MIPS of each VM are also monitored at the server level using PERF. At the same time dynamic power of server is monitored remotely using the external Power Distribution Unit “Raritan JSON-RPC(PX2)”.

V. RESULTS AND DISCUSSION

Instruction measurements observed at the VM by changing its workload are presented in Figure 2, showing the increase in number of instructions at the VM with increasing load on its vCPU. This change in instructions further increases with increase in number on VMs running on the server. More instructions per second processed by the server will require more power resources. The change in power at the server with changing load at the VMs is shown in the right part of Figure 2. The power for a single VM with increasing load changes from 0.49 W to 8.5 W, whereas this change for 2 VMs running in parallel is from 1.6 W to 13.3 W. With these observations, the VM power models are trained with two different methods. At first, LLSR is used to train data sets for model 1 and model 2. The parameters obtained for these models are listed in the top part of Table II. Similar data sets are then used in LSVR for both model 1 and model 2; model parameters obtained are listed in the bottom part of Table II.

The mapping of data points for Model1 is shown in the top left part of Figure 3. As the measured data points show an almost linear behavior, the predictions of the data points for both models are very close to the measured values. The Root

TABLE II
MODELS' PARAMETERS

LLSR				
Parameter	Model 1	Model 2 (equiv.)	Model 2a	Model 2b
β_0	0.5369	1.4395	32.769	0.1687
β_1	0.0025	0.0021	0.0533	0.0388
LSVM				
β	0.1074	-	2.8876	0.5398
ϵ	0.4030	-	10.807	0.403
C	4.0305	-	108.07	4.0305

Mean Square Error (RMSE) for both models is also small, as can be seen in Table III. Both LLSR and SVR models perform almost similarly, with RMSE of about 0.5 W. We can also observe that there is not much variation in prediction for test data in both cases as can be observed in Figure 4.

Measured and predicted values for Model2a and Model2b are shown in the lower part of Figure 3. Here we can see that the measured data for Model2a is scattered for some data points, which the SVM regression model tries to include in training, by increasing its tolerable error margin ϵ (high value of ϵ in Model2a). However, some data points are still extremely far away, which causes the cross-validation RMSE of the model to be around 44 MIPS. A similar effect was also observed for test data where the predicted RMSE value is 43 MIPS, shown in table III. The second part of Model2, i.e. Model2b, however performs much better, as shown in the bottom right corner of Figure 3. Variations in measured data points are not as many as in Model2a and the cross-validation RMSE of the model is also 1.5 W for both LLSR and LSVR (refer to Table III). The prediction error for Model2b is also small, around 1.2 W. Since Model2a and Model2b are to be used together, their combined effect is also analyzed. In the top right corner of Figure 3 combined Model2 data points are shown. It can be observed that the predicted values are overlapping the measured data points in most cases. The RMSE of test data for the combined model, Model2, is 0.6 W for LLSR and 1.07 W for LSVR, which is very much improved from individual Model 2a and Model 2b performances. However, the box-plot of prediction error for combined Model 2 in Figure 4 shows little variation as compared to Model 1, but the average prediction error still lies within the uncertainty of ± 1 W for both types of Model 2.

From the results of these models it can be concluded that Model2, through the combined parts, improves the prediction error from individual Models (2a and 2b). However, Model1 performs better than the combined Model2 when trained with LLSR. With LSVR, the combined effect in Model2 is almost similar to Model1. The percentage prediction error for Model1 is found to be 6-7%, whereas Model2's error prediction lies in the range 8-14%. The modeling approach of this paper is less complex in comparison to other existing models which use several counter values and their dependencies at the server

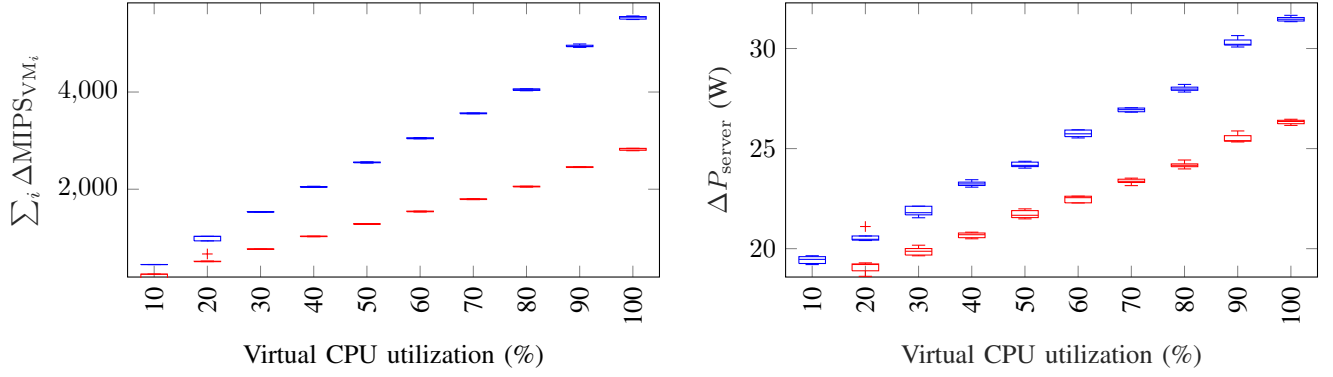


Fig. 2. Left: Increasing the vCPU utilization via a stress program leads to an increase in virtual machine MIPS. Right: The same increased vCPU utilization also increases the power consumption at the server running the VM. Legend: Only 1 VM with load (—+—), 2 VMs in parallel (—x—)

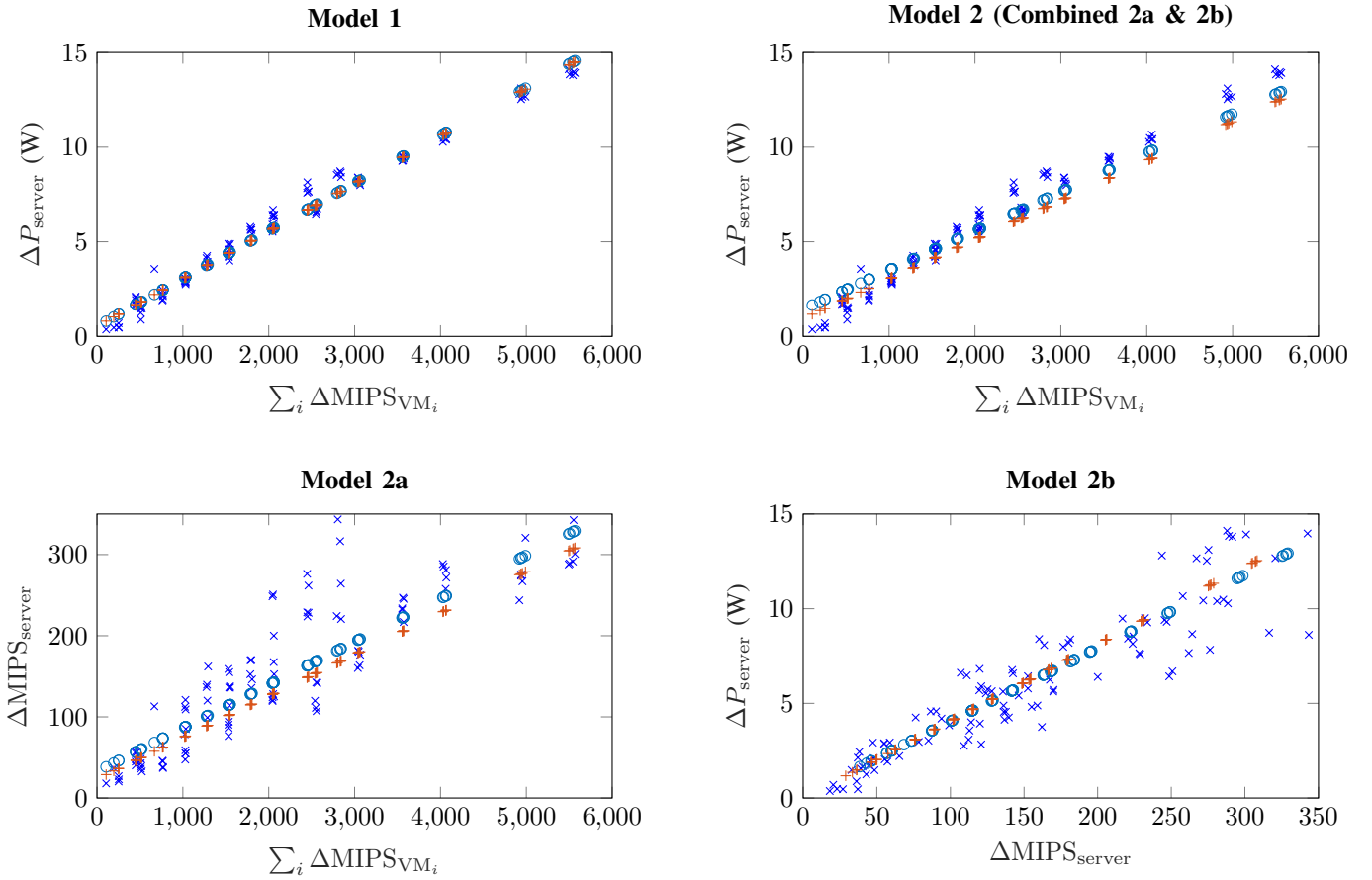


Fig. 3. Model 1 vs. 2 and individual components of model 2. Legend: “x” shows measured data used for training, while the predictions from LLSR and LSVM are shown by “o” and “+” respectively.

TABLE III
MODELS' PREDICTION ERROR

	Reg. Model	Cross-Validation RMSE	Testset RMSE
Model 1	LLSR	0.532 W	0.504 W
	LSVR	0.545 W	0.507 W
Model 2a	LLSR	44.4 MIPS	43.5 MIPS
	LSVR	45.5 MIPS	55.5 MIPS
Model 2b	LLSR	1.510 W	1.2 W
	LSVR	1.580 W	1.39 W
Model 2 (combined)	LLSR	-	0.640 W
	LSVR	-	1.070 W

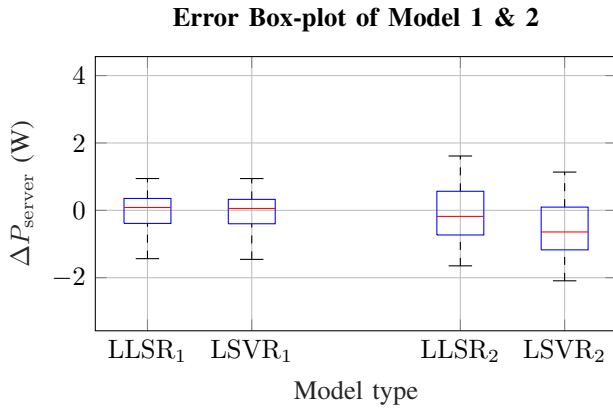


Fig. 4. Box-plot of error for Model 1 and 2. Legend: Median error (—), 1st – 3rd quartile of errors (—), overall error range (—)

level. This paper provides similar prediction accuracy (even improved prediction error in Model1), while using a less complex single counter value at the VM end.

VI. CONCLUSIONS

Measurement of power attributed to specific virtual components has become necessary and critical at the same time. With growing virtualization in servers and in the network, energy efficient load scheduling and optimized resource management is required. The estimated power consumption caused by the incoming workload to a VM can help servers making smart decisions upon VMs' initialization or their migration. This also requires precise knowledge of the actual resources consumed by individual virtual components. In this paper, both proposed models considered prediction precision along with less overhead and complexity. Model2 performs similarly to other existing models but with less complexity, whereas the accuracy in Model1 is improved (the prediction error is reduced from 10% to 6% in most cases). Further, it is expected that accuracy of both models can be improved with much larger data sets and also considering different VMs' configurations.

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